

Matías Mattamala

Curriculum Vitae

CONTACT INFORMATION

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RESEARCH OVERVIEW

I aim to make robots that help us understand and solve problems of the real world. To achieve this, my research focuses on the foundations and systems for lifelong autonomy: representations and algorithms for perception, learning, and action that enable autonomous and adaptive behaviour in robots working with humans or alone in the wild.

CURRENT POSITIONS

University of Edinburgh, Edinburgh, UK

Research Associate Jun 2025 – present

- Member of the Centre for AI for Assistive Autonomy, under Prof. Subramanian Ramamoorthy.
- Lead of the Assistive Care sub-team (5 people).
- Researching scene representations for human-robot interaction.
- Lab management, robot maintenance, student supervision (PhD, MS, UG), dissemination, public engagement.

EDUCATION

University of Oxford, Oxford, UK

DPhil in Engineering Science Oct 2019 – Nov 2023
“Vision-based Legged Robot Navigation: Localisation, Local Planning, Learning”
Supervisor: Prof. Maurice Fallon

Universidad de Chile, Santiago, Chile

M.Sc in Electrical Engineering Mar 2015 – Aug 2018
“Visual Localisation for Resource-constrained Robots”
Supervisor: Prof. Javier Ruiz-del-Solar

Ingeniería Civil Eléctrica Mar 2009 – Aug 2018
B.Sc in Electrical Engineering Mar 2009 – Mar 2014

PAST POSITIONS

University of Oxford, Oxford, UK

Postdoctoral Researcher Sep 2023 – May 2025

- Research on robot navigation and mapping using vision, geometry, and language under Prof. Maurice Fallon.
- Supervision of PhD students, student visitors, and undergraduate students.
- Lab management, grant application writing, dissemination, field trials.

ETH Zürich, Zürich, Switzerland

Visiting Researcher Apr 2022 – Sep 2022

- Research visit at the Robot Systems Lab (RSL) under Prof. Marco Hutter.
- Project on learned visual traversability estimation for natural environments.
- Co-supervision of two semester projects and two MSc theses.

Universidad de Chile, Santiago, Chile

Project Manager Jun 2018 – Aug 2019

- Coordination of the ‘Beauchef Proyecta’ unit to foster multidisciplinary activities across engineering degrees.
- Managed funds associated to the project (~£90,000).
- Supervised and co-supervised two BSc theses.

Undergraduate and MSc researcher Mar 2012 – Aug 2018

	▪ Perception lead of the UChile RoboCup Soccer team. Joined competitions in Mexico (2012), The Netherlands (2013), Brazil (2014, 4th place), and China (2015, 4th place).
	Knight Robotics, Santiago, Chile Part-time Developer Jan 2015 – Mar 2018 ▪ Assembly of educational robot kits, graphic design, training for school teachers.
	ALMA Observatory, San Pedro de Atacama, Chile Engineering Intern Jan 2013 – Mar 2013 ▪ Implementation of a GUI for monitoring the radiotelescope’s antennas.
TEACHING	University of Oxford, Oxford, UK Teaching Assistant Mar 2021 ▪ Lead TA of an AIMS CDT course on autonomous exploration. I co-designed a new online challenge (due to COVID-19 restrictions) to introduce advanced robotics methods and C++ programming.
	Universidad de Chile, Santiago, Chile Lecturer Mar 2017 – Sep 2019 ▪ I designed and taught new courses on mobile robotics(based on the Duckietown project), tech project development, battlebots, and data science for astronomy. Over 300 students taught.
	Teaching Assistant Mar 2012 – Dec 2016 ▪ TA for mobile robotics, image processing, computational methods for science and engineering, introduction to engineering, and electromagnetism. Over 200 students taught.
	Mustakis Foundation, Santiago, Chile Lecturer Apr 2014 – Jun 2018 ▪ Introductory robotics and programming courses for high school students using LEGO Mindstorms and Arduino. Over 200 high-school students taught.
AWARDS	IEEE Robotics and Automation Letters (RA-L) Honorable Mention 2025 (2026). <i>IEEE Robotics and Automation Society</i> . For Wang et al. ‘Exosense: A Vision-Based Scene Understanding System For Exoskeletons’.
	IEEE Robotics and Automation Letters (RA-L) Outstanding Reviewer (2025) – <i>IEEE Robotics and Automation Society</i> . Awarded to 5 out of 6000 reviewers for distinguished service to IEEE RA-L in 2024.
	Best Conference Paper By a Young Author (2024). <i>Intl. Work. on Robot Motion and Control (RoMoCo)</i> . For Schöneegg et al. ‘Global Path Planning for Autonomous Vehicles in Orchards and Vineyards’.
	Best Cognitive Robotics Paper Finalist (2024). <i>IEEE International Conference on Robotics and Automation</i> . For Zhang & Jin et al. ‘Resilient Legged Local Navigation: Learning to Traverse with Compromised Perception End-to-End’.
	Excellence Award (2024) – <i>Department of Engineering Science, University of Oxford</i> . In recognition of outstanding performance in 2023/2024.
GRANTS	PhD Exchange Fellowship (2022) – <i>NCCR Robotics, Swiss National Science Foundation (SNSF)</i> . £8,000 to partially support research visit at ETH Zurich.
	ANID Becas Chile Scholarship (2019 – 2023) – <i>Government of Chile</i> . Full scholarship (£145,000) for PhD studies at the University of Oxford.
	Graduate Students Project Grant (2016) – <i>Universidad de Chile</i> . £2,500 to implement a new mobile robotics course for second-year engineering students.
INVITED TALKS	How Robots Represent the World ▪ Computational Cognitive Science Group, University of Edinburgh, UK. Feb 2026.
	Robot Perception for Real-world Autonomy

- *International Workshop of Intelligent Autonomous Learning Systems (IWIALS) 2025*, Austria. Jul 2025. [\[slides\]](#)
- Universidad del Desarrollo, Chile. May 2025.
- *ICRA 2025 LA@Chile Satellite*, Universidad de Chile, Chile. May 2025.

Effective Design of Graphics for (Robotics) Research

- School of Informatics, University of Edinburgh, UK. Feb 2026. [\[slides\]](#)
- *International Workshop of Intelligent Autonomous Learning Systems (IWIALS) 2025*, Austria. July 2025. [\[slides\]](#)
- *ICRA 2025 LA@Chile Satellite*, Universidad de Chile, Chile. May 2025. [\[slides\]](#)
- Oxford Robotics Institute, University of Oxford, UK. Apr 2025. [\[slides\]](#)

Visual Navigation for Legged Robots in Challenging Environments

- *Qué hacemos en robótica?* (Spanish), Universidad de Chile, Chile. Feb 2023.
- *Técnicas modernas de control y aprendizaje de máquina para robótica y locomoción* (Spanish), Universidad Nacional de Colombia, Colombia. Feb 2023. [\[slides\]](#)
- *AIMS-WASP Event*, University of Oxford, UK. Oct 2022. [\[slides\]](#)

On Physical, Algebraic, Geometric, and Probabilistic Descriptions in Robotics

- *RPL Robotics Seminar*, University College London, UK. Oct 2021. [\[slides\]](#)

Visual Navigation for Quadrupedal Robots

- *Charlas en AI, Robótica, Tecnología y Aplicaciones* (Spanish), Universidad de O'Higgins, Chile. Jul 2021. [\[slides\]](#)
- *IEEE UPAO Webinar* (Spanish), Universidad Privada Antenor Orrego, Perú. Apr 2022. [\[slides\]](#)

KNOWLEDGE EXCHANGE

Scientific Advising

- Library of the Congress of Chile (2018 – 2021). Topic: Impact and State-of-development of Robotics in Chile. *Committee Member*
- Universidad de Chile (2018 – 2019). Topic: Innovation in Engineering Courses. *Committee Member*

Public Engagement

- Speaker: “How Robots Help Us Solve Problems in the Real World”, *William Bright Society Lecture*, Glenalmond College, UK. Mar 2026.
- Demonstrator: “Teach Me: an Interactive Robot Learning Experience”, *Edinburgh Science Festival*, Edinburgh, UK. Apr 2026. [\[news\]](#)
- Support Team: “Christmas Lecture 2023: Professor Mike Wooldridge - The Truth about AI”, *Royal Institution*, London, UK. Dec 2026.
- Speaker: “Robot Soccer”, *Futuristas, el mundo cambia contigo* (Spanish), National Congress of Chile, Chile. Nov 2016. [\[video\]](#) [\[slides\]](#)

Industry Engagement

- Speaker: “Robot Representations for Real-World Autonomy”, *Ekumen R&D Seminar*, Jul 2026.
- Speaker: “From Industrial Facilities to Forestry Applications: ANYmal Research at the University of Oxford”, *Institution of Engineering and Technology (IET)* (Invited talk by ANYbotics AG), Sep 2024. [\[video\]](#) [\[slides\]](#)
- Speaker: “DigiForest: Developing Robotic Technologies for Sustainable Forestry”, *ForestryAI Webinar*, Jul 2024. [\[video\]](#) [\[slides\]](#)

SUPERVISION AND CO-SUPERVISION

University of Edinburgh

- Heramb Modugula, PhD project (2025 – present, *Co-supervisor*). Topic: “Personalised Human-Robot Interaction”.
- Aditya Kamireddypalli, PhD project (2025 – present, *Co-supervisor*). Topic: “Visuo-Tactile Estimation for Contact-rich Manipulation”.

- Aiden Lyons, BSc(Hons) Project (2025 – 2026, *Co-supervisor*). Topic: “Force-controlled Visual-Language-Action models”.
- Tianqi Zhu, BSc(Hons) Project (2025 – 2026, *Co-supervisor*). Topic: “Contact-Rich Reinforcement Learning”.

University of Oxford

- Jin Jin, PhD project (2025 – 2026, *Co-supervisor*). Topic: “Robot Social Navigation with Foundation Models”.
- Haedam Oh, MEng project (2023, *Co-supervisor*). Topic: “Evaluation and Deployment of LiDAR-based Place Recognition in Dense Forests”.
- Leonard Freißmuth, MSc Visitor (2023, *Co-supervisor*). Topic: “Online Tree Reconstruction and Forest Inventory on a Mobile Robotic System”.
- Jianeng Wang, PhD project (2023 – 2025, *Co-supervisor*). Topic: “Vision-centric State Estimation and Mapping for Visually Challenging scenarios”.
- Christina Kassab, PhD project (2022 – 2026, *Co-supervisor*). Topic: “Unifying Semantic and Geometric Mapping with Visual Foundation Models”.

ETH Zürich

- Piotr Libera, Semester Project (2023, *Co-supervisor*). Topic: “Semantic Understanding of Outdoor Environments for Navigation”.
- Giacomo Manzoni, Semester Project (2022, *Co-supervisor*). Topic: “Motion-primitives Planning for Legged Robots”.
- Pascal Lieberherr, MSc Thesis (2022, *Co-supervisor*). Topic: “Local Path Planning in Orchards and Vineyards”.
- Timo Schöneegg, MSc Thesis (2022, *Co-supervisor*). Topic: “Global Planning in Orchards and Vineyards”.

Universidad de Chile

- Andrés Astudillo, BSc Thesis (2020, *Co-supervisor*). Topic: “Design and Construction of a Multi-purpose SCARA robot”.
- Matías Zamora, BSc Thesis (2019, *Main supervisor*). Topic: “IoT-enabled Vermicompost System”.

PROFESSIONAL ACTIVITIES

Associate Editor

- IEEE Robotics and Automation Letters (2026 – present)
- IEEE International Conference on Robotics and Automation (2026)*
- IEEE/RSJ International Conference on Intelligent Robots and Systems (2025, 2026)*

(*An *Associate Editor (AE)* at an IEEE conference has the same role of an *Area Chair (AC)* or a *Programme Committee member (PC)* in other fields)

Reviewer

I generally serve as a reviewer for:

- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Robotics (T-RO)
- IEEE Transactions on Field Robotics (T-FR)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- International Journal of Robotics Research (IJRR)
- Robotics: Science and Systems (RSS)

I have also reviewed for Nature Communications, Scientific Reports, Conference on Robot Learning (CoRL), British Machine Vision Conference (BMVC), IEEE/CVF International Conference on Computer Vision (ICCV), Scientific Reports, IEEE/CVF International Conference on Computer Vision and Pattern Recognition (CVPR).

Workshop Organisation

- *Act to Sense to Act Better: 1st Workshop on Learning and Representations for Active Perception in Manipulation*, IEEE International Conference on Robotics and Automation, Vienna, Austria (2026). *Co-organiser*
- *LocoLearn: From Bioinspired Gait Generation to Active Perception*, Conference on Robot Learning, Munich, Germany (2024). *Co-organiser*
- *Retrospective and Future of World Representations for Lifelong Robotics*, IEEE/RSJ International Conference on Intelligent Robots and Systems, Abu Dhabi, UAE (2024). *Program Committee Member*

Technical Memberships

- British Machine Vision Association (BMVA)
- Computer Vision Foundation (CVF)
- Institute of Electrical and Electronics Engineers (IEEE)

Volunteering

- 7th UK Robot Manipulation Workshop, Edinburgh, UK (2026). *Demonstrator*
- IEEE/RSJ International Conference on Intelligent Robots and Systems, Madrid, Spain (2018). *Organisation support*
- International Symposium on Robotics Research 2017, Puerto Varas, Chile (2017). *Organisation support*
- Singularity Summit Chile 2016, Santiago, Chile (2016). *Robotics Mentor*
- V Congress of the Future, National Congress of Chile, Santiago, Chile (2016). *Demonstrator*
- International Conference on Computer Vision, Santiago, Chile (2015). *Organisation support*

Matías Mattamala

Publications List

OVERVIEW

33 peer-reviewed articles. 1,092 citations. h-index 15 (as of July 2026, via [Google Scholar](#)).

In robotics, the main publication venues are peer-reviewed conferences and journals. In conferences, research articles must be novel work, which is assessed by at least two reviewers and archived in conference proceedings. The top conferences in robotics are the *IEEE International Conference on Robotics and Automation* (ICRA), the *International Conference on Intelligent Robots and Systems* (IROS), and *Robotics: Science and Systems* (RSS).

My most significant publications are highlighted with a star (★). Other articles that have been awarded or nominated are highlighted in **bold**. Equal contribution is denoted by an asterisk next to the authors's name (*).

JOURNAL ARTICLES

- [J10] Jonas Frey, **Matías Mattamala**, Hae-Won Park, Mayank Mittal, Georg Martius, Maike Osborne, Robert Sparrow and Marco Hutter. Aug. 2026. “[Advances, Challenges, and Opportunities for Legged Robots](#)”. *Sci. Robot.*
- [J9] Christina Kassab*, Hyeonjae Gil*, **Matías Mattamala**, Ayoung Kim and Maurice Fallon. July 2026. “LEXI-SG: Monocular 3D Scene Graph Mapping with Room-Guided Feed-Forward Reconstruction”. *IEEE Robot. Autom. Lett. (RA-L)*. arXiv: [2605.13741 \[cs.R0\]](#).
- [J8] Michal Staniaszek, Tobit Flatscher, Joseph Rowell, Hanlin Niu, Wenxing Liu, Yang You, Matt Gadd, **Matías Mattamala**, Alex Schutz, Daniele De Martini, Luke Pitt, Robert Skilton, Maurice Fallon and Nick Hawes. July 2025. “[AutoInspect: Towards Long-Term Autonomous Industrial Inspection](#)”. *IEEE Trans. Field Robotics*. **Invited to Special Issue**.
- [J7] **Matías Mattamala***, Jonas Frey*, Piotr Libera, Nived Chebrolu, Georg Martius, Cesar Cadena, Marco Hutter and Maurice Fallon. July 2025. “[Wild Visual Navigation: Fast Traversability Learning via Pre-Trained Models and Online Self-Supervision](#)” (★). *Autonomous Robots*. **Invited to Special Issue**.
- [J6] **Matías Mattamala**, Nived Chebrolu, Jonas Frey, Leonard Freißmuth, Haedam Oh, Benoit Casseau, Jonas Frey, Marco Hutter and Maurice Fallon. June 2025. “[Building Forest Inventories with Autonomous Legged Robots — System, Lessons, and Challenges Ahead](#)” (★). *IEEE Trans. Field Robotics*. **Invited to Special Issue**.
- [J5] Jianeng Wang, **Matías Mattamala**, Christina Kassab, Guillaume Burger, Fabio Elneceve, Lintong Zhang, Marine Petriaux and Maurice Fallon. Feb. 2025. “[Exosense: A Vision-Based Scene Understanding System For Exoskeletons](#)”. *IEEE Robot. Autom. Lett. (RA-L)*. **RA-L Honorable Mention 2025**.
- [J4] Marco Tranzatto, [...], **Matías Mattamala** et al. Apr. 2024. “[Team CERBERUS Wins the DARPA Subterranean Challenge: Technical Overview and Lessons Learned](#)”. *Field Robotics*. arXiv: [2207.04914](#).
- [J3] Yiduo Wang, Milad Ramezani, **Matías Mattamala**, Sundara Tejaswi Digumarti and Maurice Fallon. Sept. 2022. “[Strategies for Large Scale Elastic and Semantic LiDAR Reconstruction](#)”. *Robot. Auton. Syst.*
- [J2] Milad Ramezani, **Matías Mattamala** and Maurice Fallon. Apr. 2022. “[AEROS: Adaptive ROBust Least-Squares for Graph-Based SLAM](#)”. *Frontiers in Robotics and AI*.
- [J1] **Matías Mattamala**, Nived Chebrolu and Maurice Fallon. Apr. 2022. “[An Efficient Locally Reactive Controller for Safe Navigation in Visual Teach and Repeat Missions](#)” (★). *IEEE Robot. Autom. Lett. (RA-L)*.

- [C23] Aditya Kamireddypalli, João Moura, Russell Buchanan, **Matías Mattamala**, Sethu Vijayakumar and Subramanian Ramamoorthy. Oct. 2026. “ContactFusion: Stochastic Poisson Surface Maps from Visual and Contact Sensing”. *IEEE/RSJ Intl. Conf. Intell. Robots Syst. (IROS)*. arXiv: [2503.16592 \[cs.R0\]](#).
- [C22] Salvatore Esposito, **Matías Mattamala**, Daniel Rebain, Francis Xiatian Zhang, Kevin Dhaliwal, Mohsen Khadem and Subramanian Ramamoorthy. June 2026. “ROOM: A Physics-Based Continuum Robot Simulator for Photorealistic Medical Datasets Generation”. *IEEE Int. Conf. Robot. Autom. (ICRA)*.
- [C21] Christina Kassab*, Sacha Morin*, Martin Büchner*, **Matías Mattamala**, Kumaraditya Gupta, Abhinav Valada, Liam Paull and Maurice Fallon. Dec. 2025. “OpenLex3D: A Tiered Evaluation Benchmark for Open-Vocabulary 3D Scene Representations”. *Conf. on Neural Information Processing Systems (NeurIPS)*.
- [C20] Alex Schutz, Yang You, **Matías Mattamala**, Ipek Caliskanelli, Bruno Lacerda and Nick Hawes. Aug. 2025. “A Finite-State Controller Based Offline Solver for Deterministic POMDPs”. *Intl. Joint Conf. on Artificial Intelligence (IJCAI)*.
- [C19] Haedam Oh, Nived Chebrolu, **Matías Mattamala**, Leonard Freißmuth and Maurice Fallon. Oct. 2024. “Evaluation and Deployment of LiDAR-based Place Recognition in Dense Forests”. *IEEE/RSJ Intl. Conf. Intell. Robots Syst. (IROS)*.
- [C18] Benoit Casseau, Nived Chebrolu, **Matías Mattamala**, Leonard Freißmuth and Maurice Fallon. Oct. 2024. “Markerless Aerial-Terrestrial Co-Registration of Forest Point Clouds Using a Deformable Pose Graph”. *IEEE/RSJ Intl. Conf. Intell. Robots Syst. (IROS)*.
- [C17] Leonard Freißmuth, **Matías Mattamala**, Nived Chebrolu, Simon Schaefer, Stefan Leutenegger and Maurice Fallon. Oct. 2024. “Online Tree Reconstruction and Forest Inventory on a Mobile Robotic System”. *IEEE/RSJ Intl. Conf. Intell. Robots Syst. (IROS)*.
- [C16] Timo Schöneegg, Turcan Tuna, Fan Yang, Gabriel Waibel, **Matías Mattamala** and Marco Hutter. July 2024. “Global Path Planning for Autonomous Vehicles in Orchards and Vineyards”. *Intl. Work. on Robot Motion and Control (RoMoCo)*. **Best Conference Paper By a Young Author**.
- [C15] Christina Kassab, **Matías Mattamala**, Lintong Zhang and Maurice Fallon. May 2024. “Language-EXtended Indoor SLAM (LEXIS): A Versatile System for Real-time Visual Scene Understanding”. *IEEE Int. Conf. Robot. Autom. (ICRA)*.
- [C14] Chong Zhang*, Jin Jin*, Jonas Frey, Nikita Rudin, **Matías Mattamala**, César Cadena and Marco Hutter. May 2024. “Resilient Legged Local Navigation: Learning to Traverse with Compromised Perception End-to-End”. *IEEE Int. Conf. Robot. Autom. (ICRA)*. **Best Cognitive Robotics Paper Finalist**.
- [C13] Yifu Tao, Yash Bhalgat, Lanke Frank Tarimo Fu, **Matías Mattamala**, Nived Chebrolu and Maurice Fallon. May 2024. “SiLVR: Scalable Lidar-Visual Reconstruction with Neural Radiance Fields for Robotic Inspection”. *IEEE Int. Conf. Robot. Autom. (ICRA)*.
- [C12] Meher V. R. Malladi, Tiziano Guadagnino, Luca Lobefaro, **Matías Mattamala**, Holger Griess, Janine Schweier, Nived Chebrolu, Maurice Fallon, Jens Behley and Cyrill Stachniss. May 2024. “Tree Instance Segmentation and Traits Estimation for Forestry Environments Exploiting LiDAR Data Collected by Mobile Robots”. *IEEE Int. Conf. Robot. Autom. (ICRA)*.
- [C11] Gian Erni, Jonas Frey, Takahiro Miki, **Matías Mattamala** and Marco Hutter. Oct. 2023. “MEM: Multi-Modal Elevation Mapping for Robotics and Learning”. *IEEE/RSJ Intl. Conf. Intell. Robots Syst. (IROS)*.
- [C10] Jonas Frey*, **Matías Mattamala***, Nived Chebrolu, Cesar Cadena, Maurice Fallon and Marco Hutter. July 2023. “Fast Traversability Estimation for Wild Visual Navigation”. *Robotics: Science and Systems (RSS)*.

- [C9] Yiduo Wang, Milad Ramezani, **Matías Mattamala** and Maurice Fallon. Sept. 2021. “[Scalable and Elastic LiDAR Reconstruction in Complex Environments Through Spatial Analysis](#)”. *European Conference on Mobile Robotics (ECMR)*.
- [C8] **Matías Mattamala**, Milad Ramezani, Marco Camurri and Maurice Fallon. May 2021. “[Learning Camera Performance Models for Active Multi-Camera Visual Teach and Repeat](#)”. *IEEE Int. Conf. Robot. Autom. (ICRA)*.
- [C7] Milad Ramezani, Yiduo Wang, Marco Camurri, David Wisth, **Matías Mattamala** and Maurice Fallon. Oct. 2020. “[The Newer College Dataset: Handheld LiDAR, Inertial and Vision with Ground Truth](#)”. *IEEE/RSJ Intl. Conf. Intell. Robots Syst. (IROS)*.
- [C6] **Matías Mattamala**, María José Alfaro, Francisco Casado, Cristóbal Mesías, Gustavo Holmberg, Fernanda Sanchirico, Ramiro Insunza and Loreto Aguirre. 2019b. “Hackers, bandas y squads: Implementación de cursos flexibles para el desarrollo transversal de proyectos en la FCFM”. *XXXII Congreso de la Sociedad Chilena de Educación en Ingeniería (SOCHEDI)*.
- [C5] Cristopher Gómez*, **Matías Mattamala***, Tim Resink* and Javier Ruiz-del-Solar. July 2018. “[Visual SLAM-Based Localization and Navigation for Service Robots: The Pepper Case](#)”. *RoboCup 2018: Robot World Cup XXII*.
- [C4] **Matías Mattamala**, Matías Lasen, Rodrigo Chi, Andrés Caba, Miguel Patiño, Javier Larrondo and Viviana Meruane. 2018c. “Beauchef Proyecta: Implementación Curricular de Proyectos Multidisciplinarios”. *XXXI Congreso de la Sociedad Chilena de Educación en Ingeniería (SOCHEDI)*.
- [C3] **Matías Mattamala***, Gonzalo Olave*, Clayder González, Nicolás Hasbún and Javier Ruiz-del-Solar. July 2017. “[The NAO Backpack: An Open-Hardware Add-on for Fast Software Development with the NAO Robot](#)”. *RoboCup 2017: Robot World Cup XXI*.
- [C2] **Matías Mattamala**, Gonzalo Olave, Miguel Campusano, Cristopher Gómez, Luz Martínez, Pablo Estefó, Joakin Ugalde, Javier Urrutia, Felipe San Martín, Javier Carrasco, Pablo Villar and Rocío Gonzalez. 2017b. “Aprendizaje Interdisciplinario en Robótica: La Experiencia Innovadora de Duckietown Chile”. *XXX Congreso de la Sociedad Chilena de Educación en Ingeniería (SOCHEDI)*.
- [C1] **Matías Mattamala**, Constanza Villegas, José Miguel Yáñez, Pablo Cano and Javier Ruiz-del-Solar. July 2015. “[A Dynamic and Efficient Active Vision System for Humanoid Soccer Robots](#)”. *RoboCup 2015: Robot World Cup XIX*.

BOOK CHAPTERS

- [BC4] Marco Camurri, [...], **Matías Mattamala** et al. Apr. 2026. “[DigiForest: Digital Analytics and Robotics for Sustainable Forestry](#)”. *Recent Advances in Robotic Perception for Forestry*. Ed. by David Portugal. Studies in Computational Intelligence. Springer Nature Switzerland. ISBN: 978-3-032-15812-3. arXiv: [2604.14652 \[cs.R0\]](#).
- [BC3] Marco Camurri and **Matías Mattamala**. 2026g. “Leg Odometry for SLAM”. *SLAM Handbook — From Localization and Mapping to Spatial Intelligence*. Ed. by Luca Carlone, Ayoung Kim, Frank Dellaert, Timothy Barfoot and Daniel Cremers. Cambridge University Press.
- [BC2] **Matías Mattamala**. Nov. 2020. “La promesa de los robots en los tiempos del Covid-19: Desafíos y oportunidades para Chile”. *La Robótica al Servicio de la Pandemia*. Ed. by Sofía Calvo Foxley. (Spanish). Chile: Biblioteca del Congreso Nacional de Chile.
- [BC1] **Matías Mattamala**. 2019a. “¿Qué es la inteligencia artificial?” *Inteligencia Artificial y Bienestar de las Juventudes en América Latina*. Ed. by Lionel Brossi, Tomás Dodds and Ezequiel Passeron. (Spanish). Chile: LOM Ediciones.

PRE-PRINTS

- [P3] Aditya Kamireddypalli, **Matías Mattamala**, João Moura, Russell Buchanan, Sethu Vijayakumar and Subramanian Ramamoorthy. 2026f. “BayesContact: Uncertain Pose Estimation via Visuo-Tactile Proposals and Simulation-based Inference”. arXiv: [2607.16123 \[cs.R0\]](#).
- [P2] Jianeng Wang, **Matías Mattamala**, Christina Kassab, Nived Chebrolu, Guillaume Burger, Fabio Elnecave, Marine Petriaux and Maurice Fallon. 2025g. “LT-Exosense: A Vision-centric Multi-session Mapping System for Lifelong Safe Navigation of Exoskeletons”. arXiv: [2510.22164 \[cs.R0\]](#).
- [P1] Christina Kassab, **Matías Mattamala**, Sacha Morin, Martin Büchner, Abhinav Valada, Liam Paull and Maurice Fallon. Dec. 2024. “The Bare Necessities: Designing Simple, Effective Open-Vocabulary Scene Graphs”. arXiv: [2412.01539 \[cs.CV\]](#).

OTHER
CONTRIBUTIONS

- [O5] **Matías Mattamala** and Nived Chebrolu. Nov. 2023. “Tutorial on SLAM and factor graphs”. URL: https://github.com/ori-drs/slam_tutorial/ (visited on 31/10/2024).
- [O4] **Matías Mattamala**. Oct. 2023. “Designing Graphics for Scientific Publications”. URL: <https://github.com/mmattamala/paper-graphics> (visited on 31/10/2024).
- [O3] **Matías Mattamala**. Feb. 2021. “Reducing the uncertainty about the uncertainties, part 1: Linear and nonlinear”. URL: <https://gtsam.org/2021/02/23/uncertainties-part1.html> (visited on 16/09/2024).
- [O2] **Matías Mattamala**. Feb. 2021. “Reducing the uncertainty about the uncertainties, part 2: Frames and manifolds”. URL: <https://gtsam.org/2021/02/23/uncertainties-part2.html> (visited on 16/09/2024).
- [O1] **Matías Mattamala**. Feb. 2021. “Reducing the uncertainty about the uncertainties, part 3: Adjoints and covariances”. URL: <https://gtsam.org/2021/02/23/uncertainties-part3.html> (visited on 16/09/2024).